CBSE Class 10 Mathematics — Coordinate Geometry — Solutions

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1. **1.** The distance of the point $P(3, 4)$ from the origin is

   Answer: (B)

   Solution:

   1. Identify coordinates of origin as $(0, 0)$
   2. Apply distance formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
   3. Substitute values $d = \sqrt{(3 - 0)^2 + (4 - 0)^2}$
   4. Calculate $d = \sqrt{9 + 16} = \sqrt{25} = 5$

   Derivation:

   $$d = \sqrt{(3-0)^2 + (4-0)^2}\ d = \sqrt{3^2 + 4^2}\ d = \sqrt{9 + 16}\ d = \sqrt{25} = 5$$

   Coordinate Geometry — Distance Formula

2. **2.** The coordinates of the midpoint of the line segment joining the points $A(-2, 8)$ and $B(-6, -4)$ are

   Answer: (C)

   Solution:

   1. Recall midpoint formula $M = (\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$
   2. Substitute $x_1 = -2, x_2 = -6$ and $y_1 = 8, y_2 = -4$
   3. Calculate $x = \frac{-2+(-6)}{2} = \frac{-8}{2} = -4$
   4. Calculate $y = \frac{8+(-4)}{2} = \frac{4}{2} = 2$

   Derivation:

   $$M = (\frac{-2-6}{2}, \frac{8-4}{2})\ M = (\frac{-8}{2}, \frac{4}{2})\ M = (-4, 2)$$

   Coordinate Geometry — Section Formula

3. **3.** The point on the x-axis which is equidistant from points $A(2, -5)$ and $B(-2, 9)$ is

   Answer: (A)

   Solution:

   1. Let point be $P(x, 0)$
   2. Equate distances $PA^2 = PB^2$
   3. $(x - 2)^2 + (0 - (-5))^2 = (x - (-2))^2 + (0 - 9)^2$
   4. $x^2 - 4x + 4 + 25 = x^2 + 4x + 4 + 81$
   5. $-8x = 56, x = -7$

Derivation:

$(x-2)^2+25=(x+2)^2+81$ $x^2-4x+4+25=x^2+4x+4+81$ $-8x=56$ $x=-7$ $P(-7,0)$

Coordinate Geometry — Distance Formula

4. **4.** If the point $P(k,0)$ divides the line segment joining $A(2,-2)$ and $B(-7,4)$ in the ratio $1:2$, the value of k is

   Answer: (D)

   Solution:

   1. Use section formula for x-coordinate $x=\frac{mx_2+nx_1}{m+n}$
   2. Substitute $m=1, n=2, x_1=2, x_2=-7$
   3. $k=\frac{1(-7)+2(2)}{1+2}$
   4. $k=\frac{-7+4}{3}=\frac{-3}{3}=-1$

   Derivation:

   $$k=\frac{1(-7)+2(2)}{1+2}\ k=\frac{-7+4}{3}\ k=\frac{-3}{3}=-1$$

   Coordinate Geometry — Section Formula

5. **5.** The distance between the points $P(a,0)$ and $Q(0,b)$ is

   Answer: (B)

   Solution:

   1. Apply distance formula $d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$
   2. Substitute $x_1=a, y_1=0$ and $x_2=0, y_2=b$
   3. $d=\sqrt{(0-a)^2+(b-0)^2}$
   4. $d=\sqrt{(-a)^2+b^2}=\sqrt{a^2+b^2}$

   Derivation:

   $$d=\sqrt{(0-a)^2+(b-0)^2}\ d=\sqrt{(-a)^2+b^2}\ d=\sqrt{a^2+b^2}$$

   Coordinate Geometry — Distance Formula

6. **6.** Assertion (A): The distance of the point $P(3,4)$ from the origin is 5 units. Reason (R): The distance of a point (x,y) from the origin $(0,0)$ is given by $\sqrt{x^2+y^2}$.

   Answer: (A)

   Solution:

   1. Distance from origin formula is $d=\sqrt{x^2+y^2}$.
   2. Substitute $x=3, y=4$.
   3. $d=\sqrt{3^2+4^2}=\sqrt{9+16}=\sqrt{25}=5$.

   Derivation:

   $$d=\sqrt{3^2+4^2}\ d=\sqrt{9+16}\ d=\sqrt{25}=5$$

   Coordinate Geometry — Distance Formula

7. **7.** Assertion (A): The midpoint of the line segment joining $A(2,4)$ and $B(4,2)$ is $(3,3)$. Reason (R): The midpoint of a line segment with endpoints (x_1, y_1) and (x_2, y_2) is $(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$.

Answer: (A)

Solution:

1. Midpoint formula is $(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$.
2. Substitute $x_1 = 2, x_2 = 4, y_1 = 4, y_2 = 2$.
3. Midpoint $= (\frac{2+4}{2}, \frac{4+2}{2}) = (\frac{6}{2}, \frac{6}{2}) = (3,3)$.

Derivation:

$$M = (\frac{2+4}{2}, \frac{4+2}{2})\ M = (\frac{6}{2}, \frac{6}{2})\ M = (3,3)$$

Coordinate Geometry — Section Formula

8. **8.** Assertion (A): The point $(0,5)$ lies on the y-axis. Reason (R): For any point on the y-axis, the x-coordinate is zero.

Answer: (A)

Solution:

1. A point on the y-axis has the form $(0, y)$.
2. Since the given point is $(0,5)$, $x = 0$, therefore it lies on the y-axis.

Derivation:

$$\begin{aligned} P(x,y) &\text{ on } y\text{-axis} \\ \Rightarrow x = 0\ P(0,5) \\ \Rightarrow x = 0 \end{aligned}$$

Coordinate Geometry — Introduction

9. **9.** Assertion (A): The distance between the points $A(0,0)$ and $B(0,7)$ is 7 units. Reason (R): The distance between two points (x_1, y_1) and (x_2, y_2) is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

Answer: (A)

Solution:

1. Use distance formula: $\sqrt{(0-0)^2 + (7-0)^2}$.
2. $d = \sqrt{0^2 + 7^2} = \sqrt{49} = 7$.

Derivation:

$$d = \sqrt{(0-0)^2 + (7-0)^2}\ d = \sqrt{0+49}\ d = 7$$

Coordinate Geometry — Distance Formula

10. **10.** Assertion (A): The point $(3,0)$ lies on the x-axis. Reason (R): For any point on the x-axis, the y-coordinate is zero.

Answer: (A)

Solution:

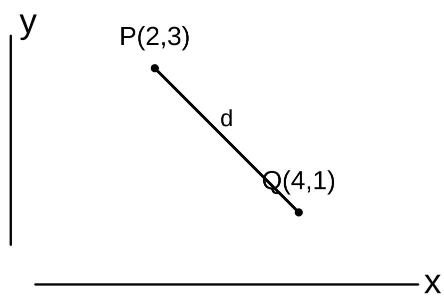
1. A point on the x-axis has the form $(x, 0)$.
2. The given point $(3, 0)$ has $y = 0$, hence it lies on the x-axis.

Derivation:

$$P(x, y) \text{ on } x\text{-axis}$$
$$\Rightarrow y = 0 \; P(3, 0)$$
$$\Rightarrow y = 0$$

Coordinate Geometry — Introduction

11. 11. Find the distance between the points $P(2, 3)$ and $Q(4, 1)$.



Answer:

$$2\sqrt{2} units$$

Solution:

1. Identify coordinates $x_1 = 2, y_1 = 3$ and $x_2 = 4, y_2 = 1$.
2. Apply distance formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

Derivation:

$$d = \sqrt{(4-2)^2 + (1-3)^2}$$
$$d = \sqrt{2^2 + (-2)^2}$$
$$d = \sqrt{4+4}$$
$$d = \sqrt{8} = 2\sqrt{2}$$

Coordinate Geometry — Distance Formula

12. 12. Determine the ratio in which the point $P(x, 2)$ divides the line segment joining the points $A(12, 5)$ and $B(4, -3)$.

Answer:

$$3 : 5$$

Solution:

1. Use section formula for y-coordinate: $y = \frac{my_2 + ny_1}{m+n}$.
2. Substitute values and solve for the ratio $m : n$.

Derivation:

$$\begin{aligned}
2 &= \frac{m(-3)+n(5)}{m+n} \\
2m+2n &= -3m+5n \\
5m &= 3n \\
\frac{m}{n} &= \frac{3}{5}
\end{aligned}$$

Coordinate Geometry — Section Formula

13. **13.** Find the coordinates of the midpoint of the line segment joining the points $A(-5,4)$ and $B(7,-8)$.

Answer:

$$(1,-2)$$

Solution:

1. Apply midpoint formula $M = (\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$.
2. Calculate coordinates by substituting the given values.

Derivation:

$$\begin{aligned}
M &= (\frac{-5+7}{2}, \frac{4-8}{2}) \\
M &= (\frac{2}{2}, \frac{-4}{2}) \\
M &= (1,-2)
\end{aligned}$$

Coordinate Geometry — Midpoint Formula

14. **14.** Find the value of k if the point $P(0,2)$ is equidistant from $A(3,k)$ and $B(k,5)$.

Answer:

$$k = 1$$

Solution:

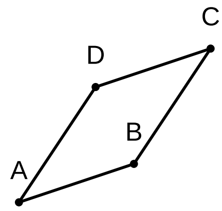
1. Use the condition $PA^2 = PB^2$ since $PA = PB$.
2. Expand and solve the resulting equation for k.

Derivation:

$$\begin{aligned}
(3-0)^2+(k-2)^2 &= (k-0)^2+(5-2)^2 \\
9+k^2-4k+4 &= k^2+9 \\
-4k+4 &= 0 \\
k &= 1
\end{aligned}$$

Coordinate Geometry — Distance Formula Application

15. **15.** If the coordinates of the vertices of a parallelogram $ABCD$ are $A(1,2)$, $B(4,3)$, $C(6,6)$, find the coordinates of the fourth vertex $D(x,y)$.



Answer:

$$(3,5)$$

Solution:

1. Use the property that diagonals of a parallelogram bisect each other.
2. Equate midpoints of AC and BD.

Derivation:

$$Midpoint of AC = (\frac{1+6}{2}, \frac{2+6}{2}) = (3.5, 4)$$
$$\frac{4+x}{2} = 3.5$$
$$\Rightarrow x = 3$$
$$\frac{3+y}{2} = 4$$
$$\Rightarrow y = 5$$

Coordinate Geometry — Properties of Quadrilaterals

16. **16.** Find the ratio in which the y-axis divides the line segment joining the points $A(5,-6)$ and $B(-1,-4)$. Also, find the point of intersection.

Answer:

$$Ratio 5:1, Point(0,-13/3)$$

Solution:

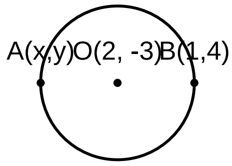
1. Assume the ratio is $k:1$. The coordinates of the point on the y-axis will be $(0,y)$.
2. Use the section formula for the x-coordinate to find k.
3. Substitute k into the section formula for the y-coordinate to find the point.

Derivation:

$$0 = \frac{k(-1) + 1(5)}{k+1}$$
$$\Rightarrow -k + 5 = 0$$
$$\Rightarrow k = 5$$
$$\text{Ratio is } 5 : 1$$
$$y = \frac{5(-4) + 1(-6)}{5+1} = \frac{-20-6}{6} = \frac{-26}{6} = -\frac{13}{3}$$
$$\text{Point is } (0, -\frac{13}{3})$$

Coordinate Geometry — Section Formula

17. **17.** Find the coordinates of a point A, where AB is the diameter of a circle whose centre is $(2, -3)$ and B is $(1, 4)$.



Answer:

$$A(3, -10)$$

Solution:

1. Let $A = (x, y)$. The centre $(2, -3)$ is the midpoint of the diameter AB.
2. Apply the midpoint formula: $x_m = \frac{x_1+x_2}{2}$ and $y_m = \frac{y_1+y_2}{2}$.
3. Solve for x and y.

Derivation:

$$\frac{x+1}{2} = 2$$
$$\Rightarrow x + 1 = 4$$
$$\Rightarrow x = 3$$
$$\frac{y+4}{2} = -3$$
$$\Rightarrow y + 4 = -6$$
$$\Rightarrow y = -10$$
$$A = (3, -10)$$

Coordinate Geometry — Midpoint Formula

18. **18.** Find the value of k for which the points $A(2,3)$, $B(4,k)$, and $C(6,-3)$ are collinear.

Answer:

$$k = 0$$

Solution:

1. For three points to be collinear, the area of the triangle formed by them must be 0.
2. Use the area formula: $\frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)| = 0$.
3. Substitute the coordinates and solve for k.

Derivation:

$$\begin{aligned} 2(k-(-3)) + 4(-3-3) + 6(3-k) &= 0 \\ 2(k+3) + 4(-6) + 18 - 6k &= 0 \\ 2k + 6 - 24 + 18 - 6k &= 0 \\ -4k &= 0 \\ \Rightarrow k &= 0 \end{aligned}$$

Coordinate Geometry — Area of Triangle

19. **19.** Find the distance between the points $P(-6,7)$ and $Q(-1,-5)$.

Answer:

$$13 units$$

Solution:

1. Use the distance formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.
2. Substitute the coordinates of P and Q.
3. Calculate the square roots to find the distance.

Derivation:

$$\begin{aligned} d &= \sqrt{(-1-(-6))^2 + (-5-7)^2} \\ d &= \sqrt{(5)^2 + (-12)^2} \\ d &= \sqrt{25+144} = \sqrt{169} \\ d &= 13 \text{ units} \end{aligned}$$

Coordinate Geometry — Distance Formula

20. **20.** If the points $A(6,1)$, $B(8,2)$, $C(9,4)$, and $D(p,3)$ are the vertices of a parallelogram taken in order, find the value of p.

Answer:

$$p = 7$$

Solution:

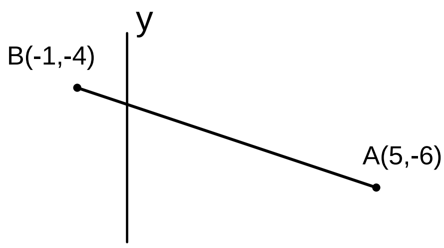
1. Diagonals of a parallelogram bisect each other, meaning their midpoints coincide.
2. Find the midpoint of diagonal AC and diagonal BD.
3. Equate the midpoints and solve for p.

Derivation:

$$\text{Midpoint of } AC = (\frac{6+9}{2}, \frac{1+4}{2}) = (7.5, 2.5)$$
$$\text{Midpoint of } BD = (\frac{8+p}{2}, \frac{2+3}{2}) = (\frac{8+p}{2}, 2.5)$$
$$\frac{8+p}{2} = 7.5$$
$$\Rightarrow 8+p = 15$$
$$\Rightarrow p = 7$$

Coordinate Geometry — Midpoint Formula

21. **21.** Find the ratio in which the y-axis divides the line segment joining the points $A(5, -6)$ and $B(-1, -4)$. Also, find the point of intersection.



Answer:

$$Ratio is 5 : 1 and the point of intersection is (0, -13/3).$$

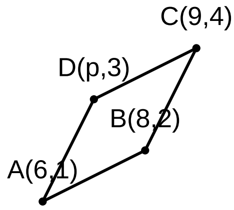
Solution:

1. Let the ratio be $k : 1$.
2. The coordinates of the point on the y-axis are $(0, y)$.
3. Use the section formula for the x-coordinate to find k.
4. Substitute k into the section formula to find the y-coordinate.
5. State the final ratio and the point.

Derivation:

$$x = \frac{m_1x_2 + m_2x_1}{m_1 + m_2}$$
$$\Rightarrow 0 = \frac{k(-1) + 1(5)}{k+1}$$
$$\Rightarrow -k + 5 = 0$$
$$\Rightarrow k = 5$$
$$\text{Ratio is } 5 : 1$$
$$y = \frac{m_1y_2 + m_2y_1}{m_1 + m_2} = \frac{5(-4) + 1(-6)}{5+1} = \frac{-20-6}{6} = \frac{-26}{6} = -\frac{13}{3}$$
$$\text{Point is } (0, -\frac{13}{3})$$

22. **22.** If the points $A(6, 1)$, $B(8, 2)$, $C(9, 4)$, and $D(p, 3)$ are the vertices of a parallelogram taken in order, find the value of p.



Answer:

$$p = 7$$

Solution:

1. Use the property that diagonals of a parallelogram bisect each other.
2. Find the midpoint of diagonal AC.
3. Find the midpoint of diagonal BD.
4. Equate the midpoints since they represent the same point.
5. Solve for p.

Derivation:

$$\begin{aligned}
\text{Midpoint of } AC &= (\frac{6+9}{2}, \frac{1+4}{2}) = (7.5, 2.5) \\
\text{Midpoint of } BD &= (\frac{8+p}{2}, \frac{2+3}{2}) = (\frac{8+p}{2}, 2.5) \\
\frac{8+p}{2} &= 7.5 \\
\Rightarrow 8+p &= 15 \\
\Rightarrow p &= 7
\end{aligned}$$

Coordinate Geometry — Midpoint Formula

23. **23.** Find the area of a triangle whose vertices are $(1, -1)$, $(-4, 6)$, and $(-3, -5)$.

Answer:

$$24 squareunits.$$

Solution:

1. Identify the coordinates $(x_1, y_1), (x_2, y_2), (x_3, y_3)$.
2. Use the area of triangle formula: $\frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$.
3. Substitute the values into the formula.
4. Perform the arithmetic calculation.

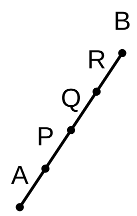
5. Express the final result in square units.

Derivation:

$$\begin{aligned}\text{Area} &= \frac{1}{2}|1(6-(-5)) + (-4)(-5-(-1)) + (-3)(-1-6)| \\ &= \frac{1}{2}|1(11) + (-4)(-4) + (-3)(-7)| \\ &= \frac{1}{2}|11+16+21| \\ &= \frac{1}{2}|48| = 24 \text{ Sq. units}\end{aligned}$$

Coordinate Geometry — Area of a Triangle

24. **24.** Find the coordinates of the points which divide the line segment joining $A(-2,2)$ and $B(2,8)$ into four equal parts.



Answer:

$$(-1,3.5),(0,5),and(1,6.5).$$

Solution:

1. Identify that four equal parts require three points of division P, Q, R.
2. Point Q is the midpoint of AB.
3. Point P is the midpoint of AQ.
4. Point R is the midpoint of QB.
5. Calculate each midpoint using the midpoint formula.

Derivation:

$$Q = (\frac{-2+2}{2}, \frac{2+8}{2}) = (0,5)$$
$$P = (\frac{-2+0}{2}, \frac{2+5}{2}) = (-1,3.5)$$
$$R = (\frac{0+2}{2}, \frac{5+8}{2}) = (1,6.5)$$

Coordinate Geometry — Section Formula